Snake with a ghost tail

mlr-agent-lab · 14 October 2026

The game

Snake in a single HTML file, `snake/index.html`: open it in a browser, no install and no build. The snake moves on a 20×20 board, one step every 110 ms. Arrow keys or WASD steer, Space pauses, Enter restarts. Eating the red apple scores a point and adds a segment; a wall ends the game. The five best scores are kept in the browser, so they survive a reload.

The twist: a ghost tail

Only the **8 newest segments** are solid. Older segments are drawn faded and are harmless: the snake can pass through them. A solid segment or a wall still ends the game. Up to 8 segments every segment is solid, so the game starts out as plain Snake; after that, each apple adds one more ghost at the tail.



Figure 1: A snake of 12 segments: the head and the next 7 are solid, the 4 oldest are ghosts. The size 8 is the constant `SOLID` in the game; the opacity 0.3 is the one it draws with.

How it was checked

The 44 tests run the page's real script in a small JavaScript engine (QuickJS) with the browser parts replaced by stand-ins, and step the game by hand. They cover movement, walls, eating, the ghost boundary, pause and restart, and the score table. **Not yet checked:** the game in a real browser (look, key handling, timing).

Rebuild this note

```
cd note && pdflatex -interaction=nonstopmode snake-note.tex
uv run --no-project --with quickjs --with pytest pytest snake -q   # the tests
```